

BigMart Retail Data Analysis - Guide

1. Introduction

Goal: Analyze product sales across stores and extract business insights.

Audience: Sales managers, business analysts, portfolio builders.

Tools: Python (Pandas, Matplotlib, Seaborn), Power BI / Tableau.

2. Understanding the Data

Columns and Description: Item_Identifier, Item_Weight, Item_Fat_Content, Item_Type, Item_MRP, Outlet_Identifier, Outlet_Establishment_Year, Outlet_Size, Outlet_Location_Type, Outlet_Type, Item_Outlet_Sales.

Analysis Goal: Identify sales patterns and relationships between features and sales.

3. Project Steps

3.1 Data Cleaning

- Remove duplicates
- Fill missing values: Item_Weight (average), Outlet_Size (mode)
- Standardize Item_Fat_Content values
- Create new column: Outlet_Age = 2025 - Outlet_Establishment_Year

3.2 Exploratory Data Analysis (EDA)

- Distribution of total sales and sales by Outlet_Type
- Sales by Item_Type and Item_Fat_Content
- Sales by Location_Type
- Scatter plot: Item_MRP vs Item_Outlet_Sales
- Correlation table between features and sales

3.3 Feature Engineering

- MRP price categories (Low / Medium / High)
- Weight categories (Light / Medium / Heavy)
- Average sales per Outlet for similar products

3.4 Dashboard / Visualization

- Power BI / Tableau: Top Selling Items, Sales per Outlet Type, Sales per Location, Trend vs Store Age
- Matplotlib/Seaborn: Similar charts in Python

3.5 Insights and Recommendations

- Identify top and underperforming stores

- Find top-selling product categories
- Highlight high-sales regions
- Provide actionable recommendations for sales and marketing strategy

4. Portfolio Suggestions

- Add a summary page with KPIs and key charts
- Include Top 10 Products and Top 5 Outlets table
- Provide practical recommendations for sales managers
- Optionally include simple trend-based sales prediction

5. Project Folder Structure

```
BigMart_Analysis/  
  data/  
    train.csv  
    cleaned_data.csv  
  notebooks/  
    Data_Cleaning.ipynb  
    EDA_Visualizations.ipynb  
  dashboard/  
    PowerBI_Dashboard.pbix  
  report/  
    BigMart_Analysis_Guide.pdf
```

6. Key Tips for Data Analysts

- Strong EDA = strong insights
- Visualizations should be clear and understandable for managers
- Keep feature engineering simple but effective
- Show all steps in dashboard or report for a professional portfolio